

# Jason Tsao

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## EDUCATION

### San Jose State University

San Jose, CA

Bachelor of Science in Computer Engineering, Minor in Japanese (GPA 3.65)

Aug. 2024 – Dec. 2027

- **Relevant Coursework:** Data Structures and Algorithms; Object-Oriented Concepts & Methodology; Circuit Analysis; Site Reliability Engineering for Cloud

## EXPERIENCE

### IT Technician III

Feb. 2026 – Present

Student Union, Inc. of SJSU

San Jose, CA

- Managed Proxmox VE, VM provisioning, backups, shutdown and high availability for reliable service continuity.
- Implemented Kubernetes for containerized workloads, enabling scalable deployments and operations for 40k users.
- Assisted with IDF infrastructure support, including drop ports, switches, patching, and ports.
- Diagnosed connectivity issues spanning SSL certificates, TCP/IP, DHCP, DNS, VLANs, NAT rules, and firewalls.
- Administered Windows and macOS endpoints, Windows Server, Active Directory, Microsoft Entra ID, and Intune.
- Managed printers, workstation deployment, imaging, software installation, and patch management.
- Secured endpoints with Qualys, Sophos, Jamf, and Intune and automated tasks with PowerShell scripts.

### Software Engineer, Development Team

Jan. 2026 – Present

Software and Computer Engineering Society

San Jose, CA

- Architected real-time YouTube control system with FastAPI WebSockets and JWT-based RBAC using SQLAlchemy, managing 3 permission tiers and eliminating unauthorized access attempts.
- Built CI/CD pipelines with GitHub Actions integrating pytest, pylint, and Docker containerization, achieving 95% test coverage and reducing deployment time from 10 minutes to under 2 minutes.
- Designed RESTful APIs with Flask-CORS and Redis session caching, improving response times by 60% and supporting 50+ concurrent users.

### Operations Supervisor

Sep. 2024 – Feb. 2026

Student Union, Inc. of SJSU

San Jose, CA

- Supervised and developed staff by monitoring performance and providing hands-on coaching as needed.
- Onboarded new hires, enforced permit compliance for organizations, and coordinated event logistics.
- Directed emergency procedures including evacuations, fire panel checks, and resolution of security incidents.

## PROJECTS

### Home Lab | Docker, Nginx, Prometheus, Grafana, Ollama, Proxmox, pfSense, TrueNAS, WireGuard

- Configured pfSense as the primary router with VLANs, custom firewall rules, and NAT policies.
- Deployed Proxmox as a bare-metal hypervisor to host, isolate, and snapshot multiple homelab VMs.
- Set up TrueNAS on Proxmox with a virtual disk RAID pool for redundant network-attached storage.
- Configured Prometheus and Grafana for real-time metrics collection, alerting, and custom dashboards.
- Self-hosted Ollama for local LLM inference behind Nginx with Cloudflare SSL over a WireGuard VPN.

### StudyGuard | Python, FastAPI, OpenCV, OpenAI Whisper, SQLAlchemy

- Won **2nd Place** at the Seed Embodied AI Hackathon for building an AI-powered focus-monitoring system.
- Built an OpenCV and GPT-4V pipeline that classifies student focus states at 10-second intervals.
- Streamed live camera feed to a FastAPI dashboard for real-time proctoring and focus visualization.
- Used a Whisper voice agent to control Reachy Mini robot gestures and update the dnsmasq allowlist.

## TECHNICAL SKILLS

**Languages:** C/C++, Assembly, JavaScript, HTML/CSS, Python, Bash/Zsh, PowerShell

**Frameworks & Libraries:** React.js, Node.js, FastAPI, SQLAlchemy

**Infrastructure & DevOps:** Docker, Proxmox, pfSense, Nginx, Prometheus, Grafana, Ollama, TCP/IP, DHCP, DNS, VLAN, WireGuard, SSL/TLS, Linux, Git, GitHub Actions, CI/CD

**Security & Endpoint Management:** Qualys, Sophos Central, Jamf, Intune, Active Directory, Windows Server

**Certifications:** Technical Support Fundamentals (Google); Vulnerability Management Detection and Response (Qualys)

**Spoken Languages:** English (Native) | Japanese (Professional Working Proficiency) | Chinese (Elementary Proficiency)